

EXCLUSIVE

IAS PRELIMS 2026

CURRENT AFFAIRS

REVISION

GS-1 (DYNAMIC)

MODULE - 22

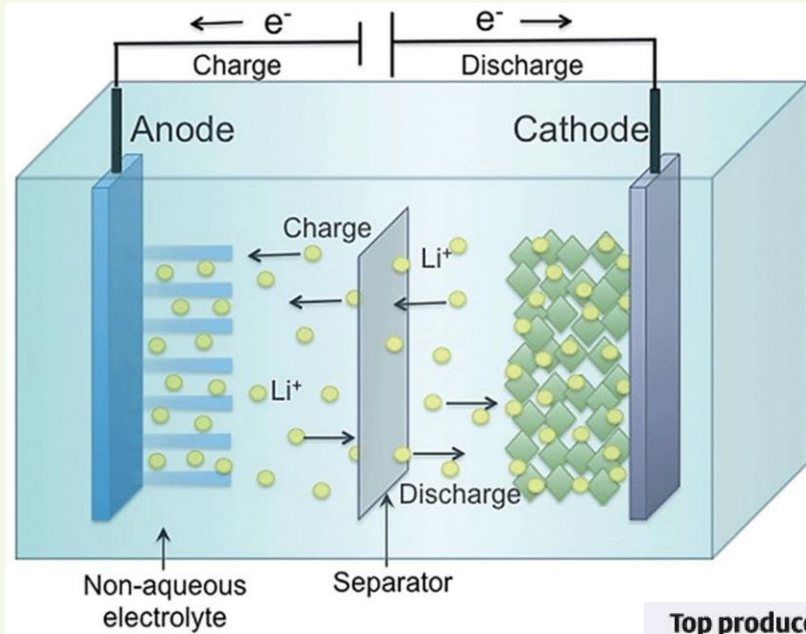


**SCIENCE
&
TECHNOLOGY-5**

What is the difference in the mechanisms between Electric Vehicles and Internal Combustion Engines? Why are lithium-ion batteries extensively used in electric vehicles?



- Electric vehicles draw electric energy from a battery to drive an electric motor and supply kinetic energy to the wheels. In an internal combustion engine, heat energy released by burning the fossil fuel to move pistons, whose motion is converted to rotary motion of the wheels.
- In the electric vehicles, lithium-ion batteries have emerged as a popular solution to storing electricity because they are energy dense, rechargeable and can be made in almost any shape.
- Lithium-ion Batteries have been known to catch fire. It is as a result of the stress creating a short circuit inside the battery. The short circuit can be due to the result of mechanical, electric and thermal abuse, which respectively deforms the internal structure, degrades its electric performance, and causes heat to accumulate.
- The batteries internal components can become corroded, due to which the risk of catching fire increases.

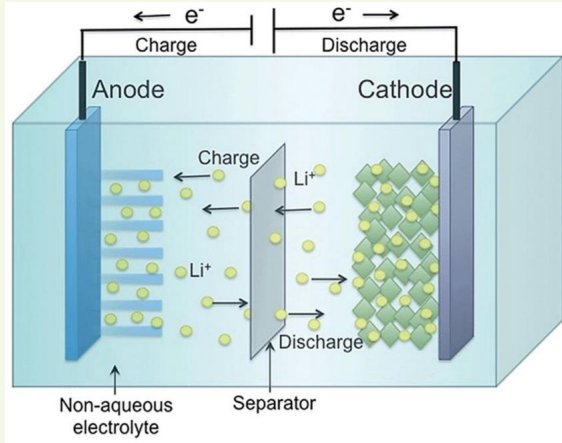


- In a cell, there are two electrodes - the anode, which is the electron giver, and the cathode, which is the electron acceptor.
- The electrons move from the anode to the cathode via an external circuit - that 'flow' which we call the electricity.
- The atoms that have lost electrons will become positively charged ions and they move from the anode to the cathode through the electrolyte.
- If the electrolyte is a liquid, the ion transfer is easy, but there are concerns of flammability.

Top producers of rare minerals 2020

Metal	Share of key producers	India's production share
Lithium	Australia (49%), Chile (22%), China (17%), Argentina (8%), Brazil (2%)	-
Graphite	China (65%), Brazil (10%), Madagascar (5%), India (3%), Mozambique (2%)	3%
Cobalt	Congo (68%), Australia (4%), Cuba (4%), Canada (3%), Philippines (3%)	-
Manganese	South Africa (32%), Gabon (16%), China (13%), Australia (10%)	5.32%
Nickel	Indonesia (33%), Philippines (13%), Russia (9%), New Caledonia (8%) Australia (7%)	-

What is the role of various components in the Lithium-Ion Batteries? Why is the electrolyte flammable?



- The anode in Lithium-ion battery is typically made up of Graphite coated on Copper Foil.
- The anode and cathode store the lithium. The battery generates electricity through chemical reactions of lithium.
- The electrolyte carries positively charged lithium ions from the anode to the cathode and vice versa through the separator.
- Cathodes, in addition to Lithium-ion, contains Nickel that delivers high energy density allowing the vehicle to travel farther.
- Cobalt is also used as it prevents the cathode from overheating, while extending the life of batteries.
- Overall cathode determines the capacity and voltage of a Lithium-ion battery. Here Manganese is also used.

In Li-ion batteries, the liquid electrolytes are organic solvents, which are highly flammable. If the current distribution is uneven, filament-like structures called dendrites can form, connecting the two electrodes and causing a short-circuit and subsequent explosions.

D. R. Congo holds more than 50% of the cobalt reserves, and Kazakhstan and South Africa together holds more than 75% of the chromium reserves

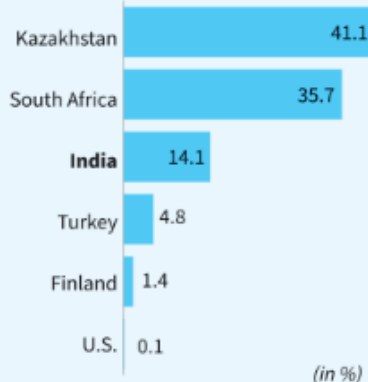
Minerals for a clean and green future

The data for the charts were sourced from Our World in Data. The data shows the reserves as a share of the global total as of 2023

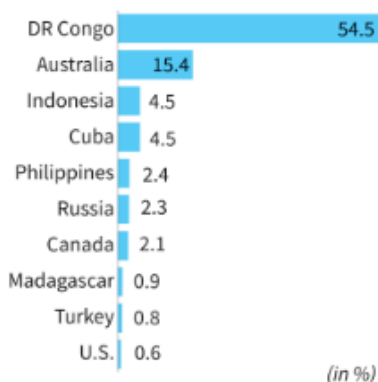
Bauxite: Guinea, Vietnam, and Australia have the highest reserves



Chromium: Kazakhstan, South Africa, and India have the highest reserves



Cobalt: Congo, Australia, and Indonesia have the highest reserves

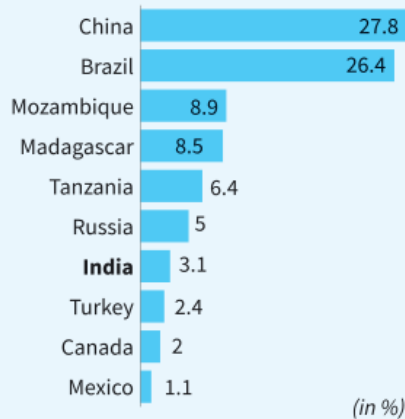


Copper: Chile, Peru, and Australia have the highest reserves

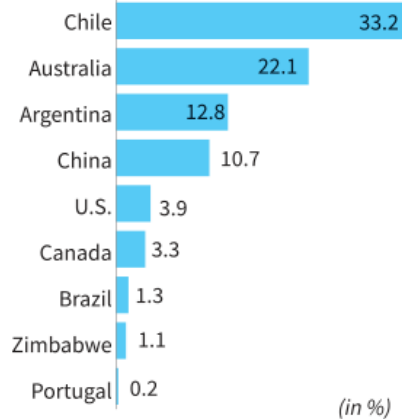


Globally, around 1/3rd of the lithium reserves are held by Chile. South Africa and Australia holds around 60% of the manganese reserves ... in molybdenum and graphite, China leads

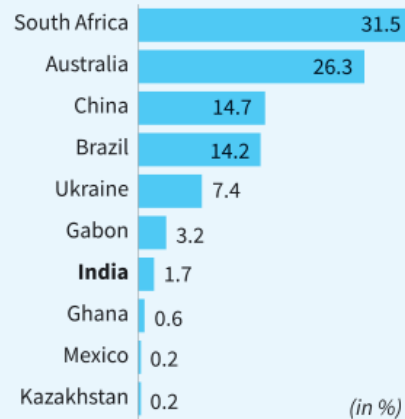
Graphite: China, Brazil, and Mozambique have the highest reserves



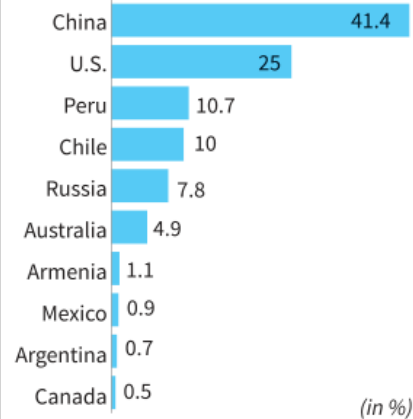
Lithium: Chile, Australia, and Argentina have the highest reserves



Manganese: South Africa, Australia, and China have the highest reserves



Molybdenum: China, U.S., and Peru have the highest reserves



China holds 40% of the Rare Earth Reserves, whereas Kazakhstan holds more than 40% of uranium reserves and Indonesia holds more than 40% of nickel reserves.





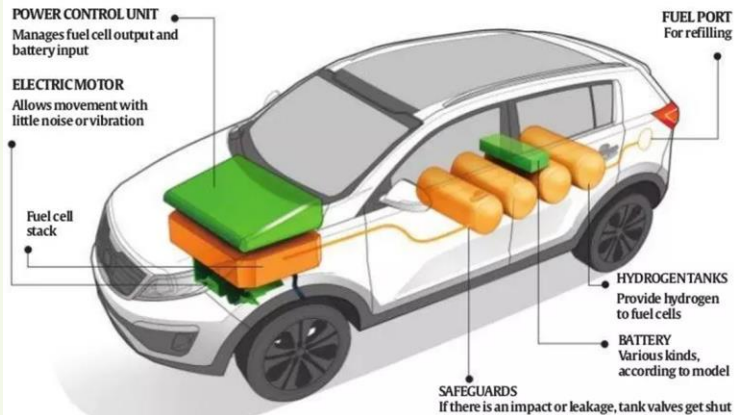
- Solid State batteries can have an energy density of 350-Watt hour per kg in comparison to 100–260-Watt hour per kg of the conventional lithium-ion batteries.
- The reason being that the absence of liquid electrolyte in the all-solid state lithium batteries simplifies the packaging and reduces the dead weight of the battery resulting in improved energy density.
- The other important reason is safety concerns, solid-state electrolytes are typically less reactive than today's liquid or gel type electrolytes and they will last longer.
- Phosphates and Oxides of Lithium (Lithium Aluminum and Germanium Phosphate or LAGP) and (Lithium Lanthanum and Zirconium Oxide or LLZO) are the most studied solid electrolyte materials.



What is the mechanism of the operation of Hydrogen Fuel Cell Electric Vehicles?

HOW IT WORKS

A fuel cell generates its own electricity through a chemical reaction between hydrogen and oxygen

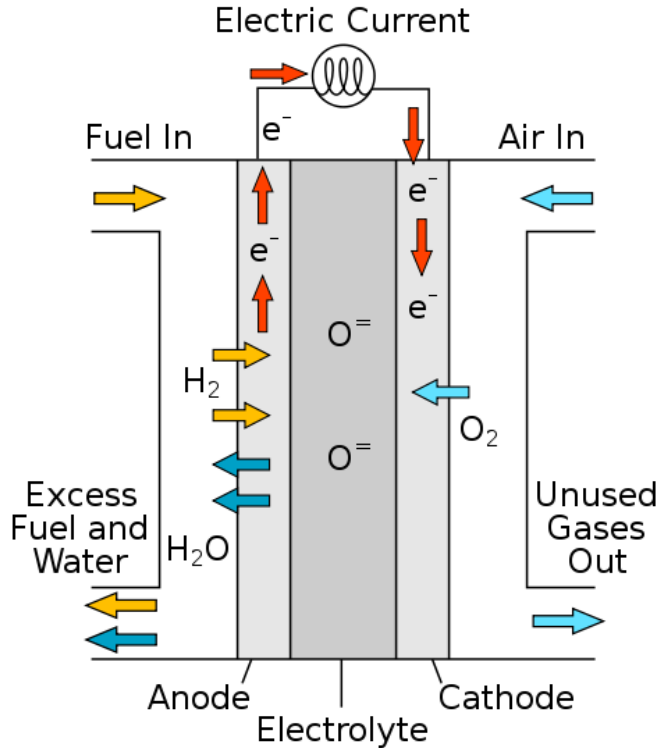


- The fuel cell combines hydrogen and oxygen to generate an electric current. Water is the by-product.
- Hydrogen fuel cells convert chemical energy into electrical energy, just like conventional batteries in the automobiles.
- FCEV vehicles are considered as vehicles of the future, given that hydrogen is the most abundant resource in the universe.
- Hydrogen fuel tanks in FCEVs are made from highly durable carbon fibre, whose strength is assessed in crash test.



India's first indigenously developed hydrogen fuel cell ferry was manufactured by Cochin Shipyard Limited (CSL) and was deployed for service at Varanasi in Uttar Pradesh. Indian Railways has taken up a state-of-the-art project for running of its first hydrogen train, on pilot basis.

Detailed Explanation about the mechanism of Hydrogen Fuel Cell Electric Vehicles?



- A hydrogen fuel cell generates its own electricity through a chemical reaction between hydrogen and oxygen.
- A fuel cell consists of anode and cathode sandwiched around an electrolyte.
- Hydrogen, is fed to the anode, and air is fed to the cathode.
- A catalyst separates hydrogen atoms into protons and electrons, which take different paths to the cathode.
- The electrons go through an external circuit, creating a flow of electricity. The protons migrate through the electrolyte to the cathode, where they reunite with oxygen and the electrons to produce water and heat.
- Power Control unit manages fuel cell output and battery input. Electric motor allows movement with little noise or vibration.

Fuel cell stack means an assemblage of fuel cells each producing electricity.

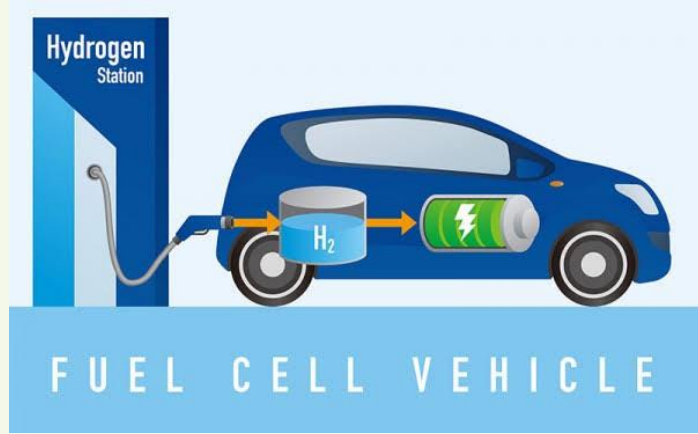
What are the advantages of using Hydrogen Fuel Cell Electric Vehicles?



- If pure hydrogen is used, fuel cells emit heat and water as a by-product.
- If Green Hydrogen is used, there is no carbon footprint.
- These are far more energy efficient than the traditional combustion technologies.
- Unlike battery powered vehicles, these need not plug in. Most models exceed a 300 km range on a full tank.

In India, Ministry of New and Renewable Energy is supporting various projects in academic institutions. 14 R&D projects on Hydrogen and Fuel cells are under implementation. The involved institutions include IITs, IISc-Bengaluru.

What are the disadvantages of using Hydrogen Fuel Cell Electric Vehicles?



- Vehicles are expensive.
- Fuel dispensing pumps are scarce.
- The process of making Hydrogen needs energy, often from fossil fuel sources. More than 90% of hydrogen is produced by steam methane reforming, which emits carbon dioxide. It can be produced by pyrolytic means using renewable feed-stocks, but that is an expensive process.
- There are questions of safety. Hydrogen is more explosive than Petrol. Opponents of the technology cite the case of hydrogen filled Hindenburg air ship incident in 1937.



IRAN WAS ORIGINALLY A SIGNATORY OF NPT

- Iran, a signatory of the Nuclear Non-Proliferation Treaty (NPT), ceased to cooperate with the International Atomic Energy Agency (IAEA) after the 1979 revolution.
- Ever since, the Islamic Republic faced allegations that it has been pursuing a clandestine nuclear programme.
- In November 2011, the IAEA reported that Iran appeared to have worked on designing an atom bomb.
- Subsequently, in 2015 Iran nuclear deal (JCPOA) was signed.

WHAT ARE THE CONTENTS OF THE JOINT COMPREHENSIVE PLAN OF ACTION (JCPOA)?



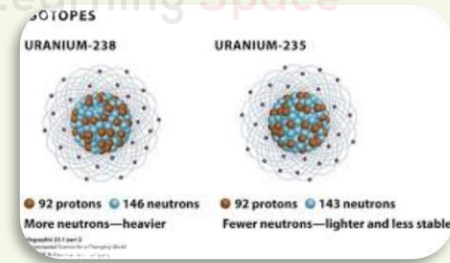
- The 2015 Iran nuclear deal is known as the Joint Comprehensive Plan Of Action (JCPOA). This is between Iran, the five permanent members of the UN Security Council and the EU.
- It promised to remove international sanctions on Iran in return for the country removing most of its centrifuges, limiting enrichment to 3.67%, and capping its LEU stockpile at 300 kg, among other measures.
- As per experts, Iran was fully compliant with the terms when Mr. Trump pulled the US out of it in May 2018 and reimposed sanctions on Tehran. Iran has since accelerated its nuclear programme breaching the agreement, which saw the country enriching uranium to 60% at its plants.

WHAT DO YOU UNDERSTAND BY URANIUM ENRICHMENT?

- In natural settings, **U-235**, the **uranium isotope** that can **sustain nuclear fission chain reactions** makes up just around **0.7% of uranium**. The rest is U-238.
- Before its use in nuclear settings, **uranium** is **enriched** to **increase the concentration of U-235**.
- Both Low-enriched Uranium (**LEU**) and High-Assay Low-Enriched Uranium (**HALEU**) are **enriched** up to **20%** for use in different kinds of nuclear reactors.
- Highly Enriched Uranium (**HEU**) **refers** to **enrichment beyond 20%**. Weapons-grade uranium is typically 90% or more.

CENTRIFUGES ARE THE WORLD'S ENRICHMENT TECHNOLOGY OF CHOICE

- **Centrifuges** are the **containers** that **spin** their contents at **several thousand revolutions** per minute.
- As **U-238** is slightly denser than **U-235**, the **centrifugal force** **pushes** it to **more towards** the **periphery**. The feed is **Uranium Hexafluoride (UF₆)** gas.



THE PRESENT STATE OF NUCLEAR POWER PRODUCTION IN INDIA

- The Government targets 100 GW by 2047. The interim target for 2031-32 is 22,480 MW.
- India has so far built only 8,180 MW of nuclear power capacity, of which 2,000 MW has been built with Russian reactors. Hence, there is huge task before the government.

THE GOVERNMENT AMENDED TWO ACTS

- Government amended the Atomic Energy Act and Civil Liability for Nuclear Damage Act (CLNDA). Amending the Atomic Energy Act is to let in private players.
- The government addressed the key clause under the CLNDA that has deterred global private nuclear players. CLNDA imposed liabilities on suppliers as well as operators in the case of nuclear damage.

CORE-LOADING EVENT AT PFBR

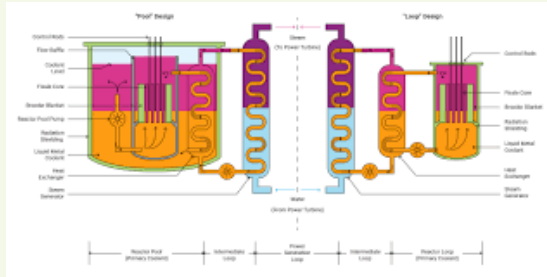
- On March 4th, 2025, Prime Minister witnessed the start of the process of **core-loading** of the indigenous Prototype Fast Breeder Reactor (**PFBR**) at the **Madras Atomic Power Station** in Kalpakkam, Tamil Nadu. It is being administered by BHAVINI.
- The **PFBR** is a **machine** that **produces** more **nuclear fuel** than it consumes.
- Its core-loading event is being hailed as a milestone because the **operationalization** of the **PFBR** will mark the **start** of **stage II** of India's **three-stage nuclear power programme**.



WHAT DO YOU UNDERSTAND BY BHARATIYA NABHIKIYA VIDYUT NIGAM LIMITED (BHAVINI)?

- It is a wholly owned enterprise of Government of India under the administrative control of the Department of Atomic Energy. It is a **public limited company** under the **Companies Act, 1956**. It was incorporated in 2003.
- The objective is constructing and commissioning the first **500 Mega Watt (Mwe) Fast Breeder Reactor (FBR)** at **Kalpakkam** in **Tamil Nadu** and to pursue construction, **commissioning, operation** and **maintenance** of subsequent **fast breeder reactors** for generation of electricity, once the first fast breeder reactor called Prototype Fast Breeder Reactor goes into commercial power production.
- **BHAVINI** will be the **second power utility** in India after Nuclear Power Corporation of India (**NPCI**) to use **nuclear fuel sources** to **generate power**.
- The **500 MW** Prototype Fast Breeder Reactor (**PFBR**) has been **under construction** **since 2002**.

What do you understand by Fast Breeder Reactors (Fast Neutron Reactors)?



- Fast breeder reactors are a class of nuclear power plants that are designed to produce energy from nuclear fission but also breed more fuel than they consume for the fission reaction.
- As the atoms of fissile material (usually, Uranium-235) split to produce heat, the neutrons that escape from the fission reaction bombard a blanket of naturally available U-238, making another nuclear fuel called Plutonium-239.
- Since these reactors produce more Plutonium-239 than the Uranium-235 they consume, they are called breeder reactors.
- It is also possible to have a blanket of Thorium, a fertile material (that India has abundance of), so that the Thorium mutates into a fissile material, Uranium-233. India has the world's largest reserves of Thorium i.e., 1.07 million tonnes.
- The World gave up the once-popular breeder reactors decades ago, as enough Uranium was discovered and mined obviating the need to go through the cumbersome process of making Plutonium in a costlier breeder reactor.



1

STAGE- I



- In the first stage, India used Pressurized Heavy Water Reactors (PHWRs) and natural Uranium -238 (U-238), which contains miniscule amounts of U-235 as the fissile material. In the process of nuclear fission, the nucleus of an atom absorbs a neutron, destabilizes and breaks into two while releasing some energy.
- If the destabilized nucleus releases more neutrons, the reactors will attempt to use them to instigate more fission reactions. The heavy water in PHWR slows neutrons released by one fission reaction enough to be captured by other U-238 and U-235 nuclei and cause new fission. The heavy water is pressurized to keep it from boiling.
- The reactions produce Plutonium-239 (Pu-239) and energy.

Pressurized heavy water reactor is nothing but water molecules containing the deuterium isotope of hydrogen.

INDIA'S PLANNED THREE-STAGE NUCLEAR POWER PROGRAM

Learning Space

2

STAGE-II

- Only U-235, not U-238, can sustain a chain reaction but it is consumed fully in stage I.
- In stage II, India will use Pu-239 together with U-238 in the PFBR to produce energy, U-233, and more Pu-239.
- The Department of Atomic Energy set up a Special-purpose Vehicle in 2003 called Bharatiya Nabhikiya Vidyut Nigam Limited (BHAVINI) to implement stage II.

3

STAGE-III

- Pu-239 will be combined with Thorium-232 (Th-232) in reactors to produce energy and U-233.
- Homi J. Bhabha designed the three-stage programme because India hosts roughly a quarter of the world's thorium reserves.

The three stages are expected to allow the country complete self-sufficiency in nuclear energy.

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